320654(20)

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BE (6th Semester)

Examination, April - May, 2016

(New Scheme)

Concrete Technology

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Attempt **all** questions, selecting any two parts from each (*b*), (*c*) and (*d*). Part (*a*) is compulsory. Wherever needed use graph drawn on answer copy only to support your answers. No any CODES are permitted.

(*ii*) The figures in the right-hand margin indicate marks.

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1. (*a*) What is pozzolonic reaction? 2

(*b*) What is the significance of following properties of aggregates? 7

(*i*) Soundness

(*ii*) Shape

(*iii*) Gap graded aggregates

(*iv*) Ten percent finer value

(*c*) (*i*) Discuss in brief the mineral characteristics and thermal characteristics of aggregates. 3

(*ii*) Explain surface index method of combining aggregates. 4

(*d*) Give graphical representations of structure of hydrated cement paste, capillary pores at different age and water cement ratio. 7

2. (*a*) How are the voids caused in fresh concrete? 2

(*b*) Give following descriptions regarding workability of concrete : 7

(*i*) Factors affecting workability

(*ii*) Causes of bleeding, segregation and crazing of concrete. 7

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(*c*) Discuss in brief the principles of measurement of workability by different test. What are the limitations you see in measuring workability by above tests? 7

(*d*) Write short notes on uses of flyash and silica fume in concrete as mineral admixtures. 7

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3. (*a*) Define capillary pores. What are its importance in relation to hardened concrete? 2

(*b*) Explaining Abram's Law, Feret's Law and Power's Law, discuss effect of water-cement ratio on compressive strength of concrete. 7

(*c*) The strength of a sample of fully maturated concrete is 40 N/mm^2. Find the strength of identical concrete at the age of 50% maturity by maturity concept. 7

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(*d*) Discuss the factors causing plastic shrinking autogenous shrinkage, drying shrinkage and carbonation shrinkage. 7

4. (*a*) Define briefly the meaning of various types of exposure condition as stated by IS 456 : 2000. 2

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(*b*) What are the principles of concrete mix design ? 7

(*c*) Design a concrete mix for heavily reinforced section, using IS Code 10262 : 2009 guidelines. Assume other conditions yourself. 7

(*d*) What is the need of Non-Destructive Testing? Discuss applications of Rebound hammer test. 7

5. (*a*) What is the need of developing special concrete? 2

(*b*) Discuss the properties of following special concrete : 7

(*i*) Light weight concrete

(*ii*) Self-compacted concrete

(*c*) Explain fiber reinforced concrete, its properties, materials required and application. 7

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(*d*) Define and discuss the following types of concrete :

(*i*) Ready mix concrete

(*ii*) Vacuum dewatered concrete

(*iii*) Tremie concrete (for underwater) 7

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